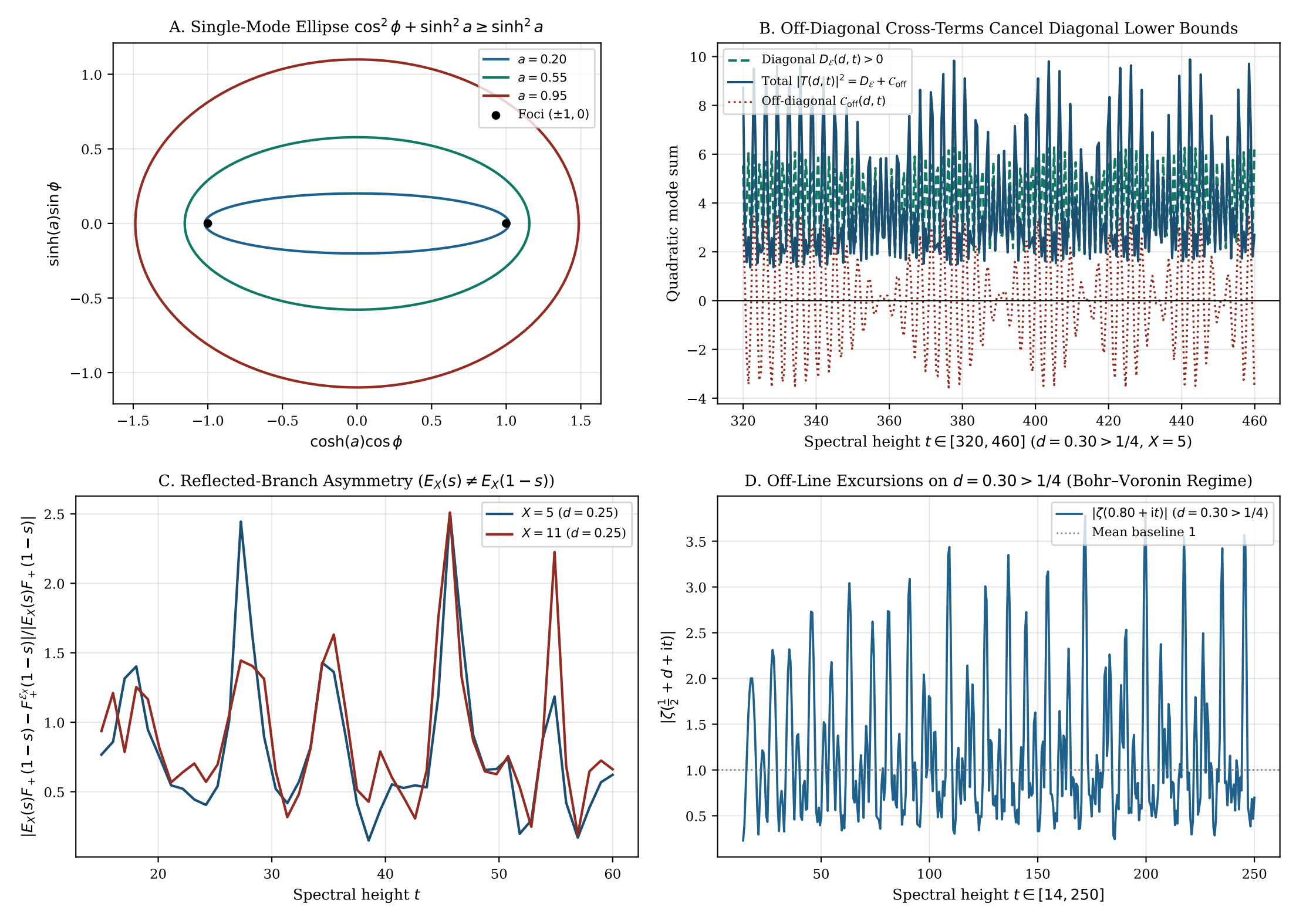

A. Single-Mode Ellipse $\cos^2\phi + \sinh^2 a \geq \sinh^2 a$
$a = 0.20$
$a = 0.55$
$a = 0.95$
Foci $(\pm 1, 0)$
$\sinh(a)\sin\phi$
$\cosh(a)\cos\phi$
B. Off-Diagonal Cross-Terms Cancel Diagonal Lower Bounds
Diagonal $D_\varepsilon(d,t) > 0$
Total $|T(d,t)|^2 = D_\varepsilon + C_{\text{off}}$
Off-diagonal $C_{\text{off}}(d,t)$
Quadratic mode sum
Spectral height $t \in [320, 460]$ $(d = 0.30 > 1/4,\ X = 5)$
C. Reflected-Branch Asymmetry ($E_X(s) \neq E_X(1-s)$)
$X = 5$ $(d = 0.25)$
$X = 11$ $(d = 0.25)$
$|E_X(s)F_+(1-s) - F_+^{\mathcal{E}_X}(1-s)|/|E_X(s)F_+(1-s)|$
Spectral height $t$
D. Off-Line Excursions on $d = 0.30 > 1/4$ (Bohr–Voronin Regime)
$|\zeta(0.80 + it)|$ $(d = 0.30 > 1/4)$
Mean baseline 1
$|\zeta(\frac{1}{2} + d + it)|$
Spectral height $t \in [14, 250]$